Sakshi Maharana (Decision Scientist IV)

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Professional Summary:

Decision Scientist with 4+ years of experience in Pharma and Retail, holding a Master's in Machine Learning & AI from IIIT Bangalore and LJMU, with expertise in analytics, prompt engineering, predictive modelling, and data visualization. Experienced in stakeholder collaboration, model governance, and supporting teams to deliver impactful, data-driven outcomes

Work Experience (Overall: 4 Years 9 months)

Mu Sigma Inc, Bengaluru, India (Tenure: 3.2 years)

Role: Decision Scientist IV — *Jan'25 – Present*

- Developed an **LLM**-based taxonomy classification pipeline using **AutoGen** and **LLaMA**, achieving **85%** accuracy for clinical trial data integration into a **knowledge graph**, enhancing entity alignment and data reliability.
- Built a prioritization framework to rank HCPs/PIs by outreach potential using clustering (K-means, GMM), improving identification of high-potential HCPs by **~41%** over legacy methods.
- Built an **LLM** pipeline to extract efficacy metrics from **100+ clinical trial PDFs** daily, improving protocol design success rates by **20%** using **Prompt Engineering** leading operational efficiency by **30%.**

Role: Decision Scientist III — *Jan'24 – Dec'24*

- Designed an **NLP-powered Python Dash** tool to analyze 500+ clinical trials, improving protocol design efficiency by **35%**, resulting in **15%** faster clinical trial initiation and educing support workload by 80% through an **enhanced Plotly Dash UI.**
- Developed an NLP pipeline to extract trial outcomes, reducing review time from 1 hr to 5 min per study with ~90% accuracy.

Role: Decision Scientist II — *Oct'22 – Dec'23*

- Developed a Principal Investigator (PI) ranking framework using KPIs across 100+ trials, **improving data quality by 30%** and enabling data-driven investigator benchmarking.
- Built an **anomaly detection pipeline** using time series, variance, and outlier analysis to monitor 150+ clinical research sites for irregular behavior.

Merkle Inc, Bengaluru, India (Tenure: 1.7 years)

Role: Software Engineer — *Apr'22 –Oct'22*

- Developed Python scripts to automate product listing, product page, and category page generation for **200+ European retail sites**.
- Improved data attribution processes by 30% through **Data Mining and crawling techniques** integrated into custom Python tooling.
- Led development efforts for web analytics and automation pipelines, increasing delivery speed and accuracy.

Role: Analyst — *Mar'21 –Apr'22*

- Built and maintained web scraping and data crawling scripts using **Python, Selenium, and BeautifulSoup** for product data pipelines.
- Improved **scraping efficiency by 20%** through optimized crawling logic and enhanced data quality control.
- Gained expertise in web analytics and structured data extraction across large-scale retail websites.

Technical Skills:

- **Tools/Languages:** Python, PyDash, SQL, LLM, RAG, Knowledge Graph, CrewAI, LangChain, LangGraph, GLiNER, OpenAI, HTML, CSS, C++, Java, AWS (S3), MS Excel
- **Tech Stack:** Matplotlib, Seaborn, TensorFlow, Keras, Sci-kit-Learn, OpenCV, YoloV3, Jupyter, Cursor IDE, Spyder, PyCharm, VSCode C++ 2019, Google Colab, UV Package Manager

Core Skills:

- Machine Learning, Natural Language Processing, Data Science, Large Language Model, Artificial Intelligence, Data Visualization, Web Scraping and Crawling, Python Dash, Predictive Analysis, Generative AI, Agentic AI Tools, Multi-Agentic System

Education:

- Master of Science in Machine Learning and AI – Liverpool John Moores University, UK — Mar'20 - Jul'22
- Post-Graduation Diploma in Data Science, IIIT Bangalore — Mar'20 - Apr'21
- B.E (Electronics and Telecommunication), Bhilai Institute of Technology, Durg — May'15 - Jun'19

Awards & Achievements:

- Agentic AI Engineering Course Completion (09/2025)
- Received Spot Award at Mu Sigma (10/2024)
- Received Champion Award at Merkle Inc (10/2021)